Restaurant Online Ordering & Table Reservation System

Progress Documentation - Activities 3, 4, 5 and 6

COMP 018 - Database Administration · Prof. John Kenneth Timblaco · Group **Binary Sisters** · Group no. __ [TO COMPLETE]

> **How to use this document.** Everything that is design or code in the repository is already written here. Items marked **[TO COMPLETE]** need real evidence that only the group can produce (screenshots, dates, commit hashes, results of testing other groups' projects). Do not invent results. Delete these notes before submitting, and check every screenshot for passwords.

1. Cover and team

Item	Value
Project title	Restaurant Online Ordering & Table Reservation System
Group	Binary Sisters (Group no. __ [TO COMPLETE])
Repository URL (access type)	<REPOSITORY_URL> (public / private with all groups invited) [TO COMPLETE]
Class tracker link	<TRACKER_URL> [TO COMPLETE]
User roles document	docs/user-roles.md in the repository (link in tracker) [TO COMPLETE]
Approved design	Activity 2 - submitted 2026 September 26

Member	Role in the project	Main contribution [TO COMPLETE]
Sandoval, Cellyn S.	[TO COMPLETE]	
Callos, Kathlen Mae L.	[TO COMPLETE]	
Romingquit, Arika E.	[TO COMPLETE]	
Tambio, Jenny Mae N.	[TO COMPLETE]	
Tungol, Jaimie N.	[TO COMPLETE]	
Gordonas, John Benedick	[TO COMPLETE]	

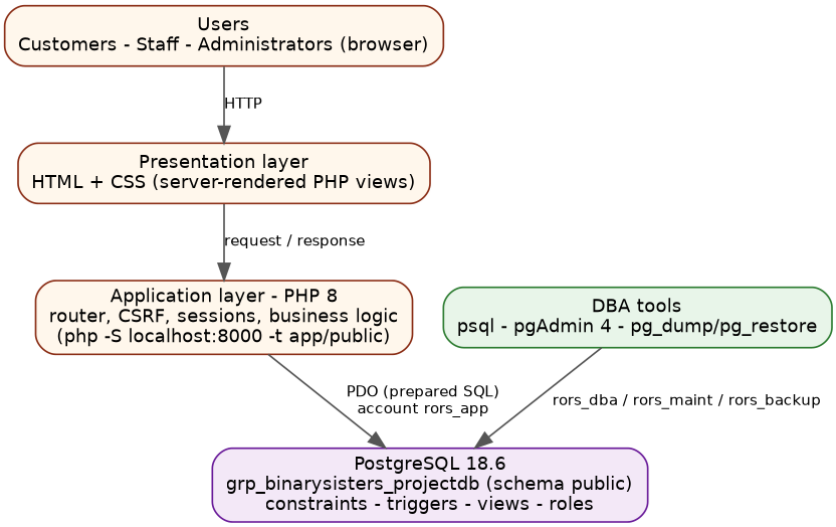
2. System overview (Activity 3)

Purpose. Restaurants that take orders and reservations by phone, message or paper suffer wrong orders, double bookings and slow status updates (Activity 2, Part 2). The system lets customers view the menu, order online, reserve a table and track both, while staff manage the operation. **Scope** follows Activity 2: payroll, suppliers, accounting and inventory are excluded.

#	Approved feature (Activity 2, Part 2B)	Where implemented
1	Customer registration and login	`/register`, `/login`, `/profile`
2-3	View menu, search and item details	`/menu`, `/menu/item`
4-5	Place orders with multiple items	`/cart`, `/checkout` (one transaction: orders + order_item + payment)
6-7	Make reservation, check availability	`/reservations/new` (availability query + database no-overlap constraint)
8	View order and reservation status	`/orders`, `/reservations`
9-10	Manage menu items and tables	`/admin/menu`, `/admin/tables`
11	Manage customer orders	`/admin/orders`

#	Approved feature (Activity 2, Part 2B)	Where implemented
12	Record and manage payments	`/admin/payments`

Architecture (unchanged from Activity 2): Users → HTML/CSS → PHP 8 application → PostgreSQL 18.6 via PDO.



System architecture

3. User roles

Full text: `docs/user-roles.md` and `docs/user-role-setup.md`.

Application role	Functions and data access	Database role / account
Customer	register, edit own profile, order, reserve, view/cancel own pending orders and reservations	via application runtime `rors_app` ∈ `rors_runtime_grp`
Staff	read operations; update order, reservation, payment status; record payments; toggle menu availability	via `rors_app` (restrictions enforced in the application)
Administrator	staff functions + create/edit/delete menu items and tables, delete reservations, customers list, staff accounts	via `rors_app`; DBA tasks use a separate account
Application runtime	only the DML the code needs; no DDL, no DELETE on orders/payments/order lines	`rors_app` ∈ `rors_runtime_grp`
Viewer / auditor	read-only on approved tables and views; no access to password hashes	`rors_report` ∈ `rors_readonly_grp`
DBA - maintenance	VACUUM/ANALYZE, monitoring	`rors_maint` ∈ `rors_maint_grp` (`pg_maintain`, `pg_monitor`)
DBA - backup	pg_dump	`rors_backup` ∈ `rors_backup_grp` (`pg_read_all_data`)
DBA - administrator	day-to-day administration, not a superuser	`rors_dba` ∈ `rors_admin_grp`

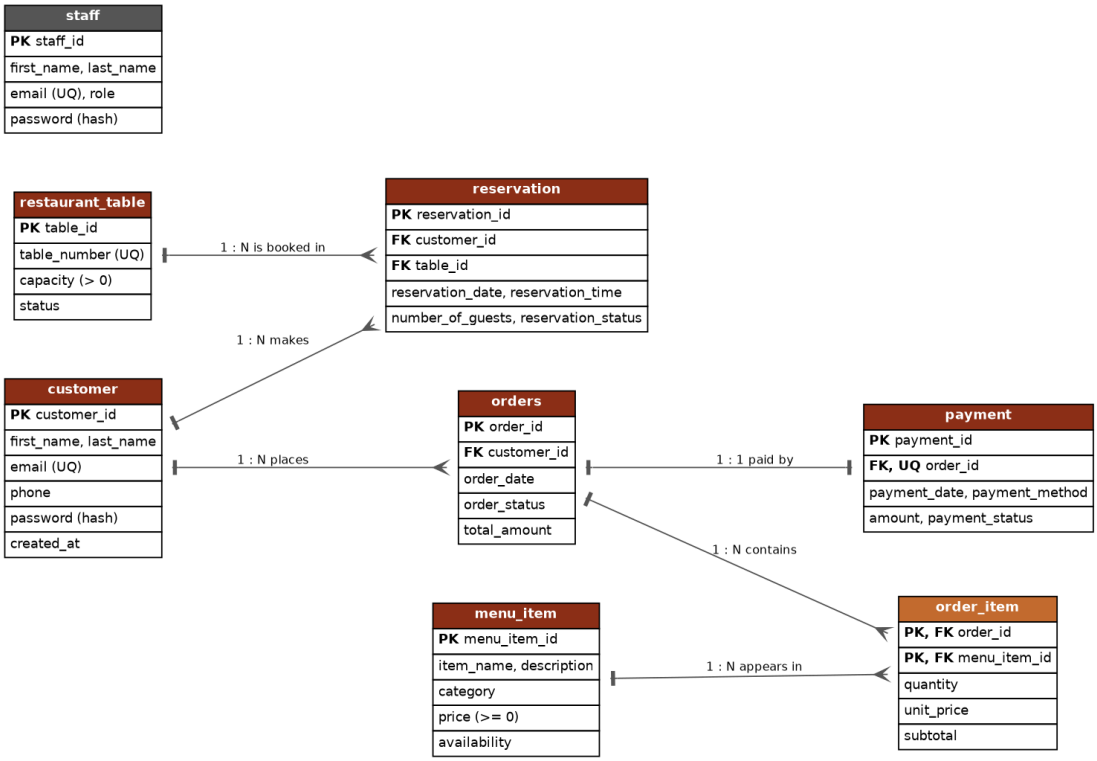
4. Development progress log

Date	Milestone / work done	Who	Key commits
2026-09-26	Activity 2 design submitted: requirements, 12 business rules, ERD, schema, architecture	all	-

Date	Milestone / work done	Who	Key commits
[date]	Repository created, members added as collaborators	[name]	[hash]
[date]	Migrations 001-004 created and run on PostgreSQL 18.6	[name]	[hash]
[date]	Seed data loaded	[name]	[hash]
[date]	PHP application: authentication and menu	[name]	[hash]
[date]	Orders, cart, checkout transaction	[name]	[hash]
[date]	Reservations and availability	[name]	[hash]
[date]	Staff/admin pages	[name]	[hash]
[date]	Roles and privileges (database/roles)	[name]	[hash]
[date]	Documentation, tracker registration, cross-group testing	all	[hash]

[TO COMPLETE - dates, names and commit hashes must come from `git log --oneline --author=...`; every member should appear.]

5. Database implementation



ERD

Item	Value
Database / schema / version / owner	grp_binarysisters_projectdb / public / PostgreSQL 18.6 / postgres
Tables (8)	customer, menu_item, restaurant_table, staff, orders, order_item, reservation, payment
Scripts	`database/migrations/000-004`, `database/seeds/001_seed_data.sql`, `database/roles/01-04`, `database/setup.ps1`
Seed volume	2 staff, 5 customers, 10 menu items, 8 tables, 6 orders (14

Item	Value
	lines), 5 payments, 7 reservations
Extra database logic	deferred triggers (order needs ≥ 1 item), total trigger, capacity trigger, exclusion constraint (no double booking), 4 views, 6 indexes

Changes from the approved Activity 2 design (details in `docs/changes-from-activity2.md`)

Change	Reason
Singular names `menu_item`, `order_item`	Activity 2 mixed singular/plural; the pgAdmin-generated ERD used singular
`price >= 0` and no `UNIQUE(item_name)`	Part 6 schema is authoritative; the Part 11 sample differed
CHECK constraints on status/role/method columns	prevent typos that break the application
Order total and subtotal consistency (trigger + CHECK)	the total must always match the lines
BR9 “order has at least one item” via deferred trigger	cannot be expressed as a column constraint
Exclusion constraint and capacity trigger	the problem statement names double bookings
Views and indexes added	planned “other objects” in Activity 2; needed for read-only access (Activity 4)

6. Application evidence (Activity 3)

Database-level test: `tests/crud_tests.sql` (runs inside a transaction that is rolled back).

Operation	Test	Expected	Actual [TO COMPLETE]	Screenshot
Create	register a customer; place an order with 2 items; reserve a table	rows created; order total = sum of lines		[Figure]
Retrieve	menu search “adobo”; “My orders”; admin order list	correct rows only		[Figure]
Update	staff sets order *confirmed*; admin edits a menu price	status/price changed; old order lines keep old price		[Figure]
Delete	admin deletes an unused menu item; deletes an item used in orders	first succeeds; second refused (foreign key)		[Figure]
Rules	book same table twice; 5 guests at a 2-seat table; negative price	all rejected		[Figure]

[SCREENSHOT REQUIRED] home/menu page · cart and order confirmation · reservation availability page · admin dashboard · `crud_tests.sql` output (all PASS).

7. Setup and configuration

1. `git clone \<REPOSITORY_URL\>`; PostgreSQL 18 + PHP 8.1+ (`pdo_pgsql`) + `psql` on PATH.
2. `.\database\setup.ps1 -WithRoles` (hidden password prompts; nothing stored).
3. `copy .env.example .env` → set `DB_USER=rors_app` and its password locally.
4. `php -S localhost:8000 -t app/public`.

Environment variable	Placeholder value
DB_HOST / DB_PORT / DB_NAME	localhost / 5432 / grp_binarysisters_projectdb
DB_USER / DB_PASSWORD	rors_app / CHANGE_ME
APP_DEBUG / APP_TIMEZONE / DB_SSLMODE	false / Asia/Manila / prefer

Database users and roles: see section 3 and `docs/user-role-setup.md`.

8. Issues and resolutions

#	Problem met	Cause	How it was solved	Who
1	[TO COMPLETE - real problems only]			

Known design-level issues found while implementing the Activity 2 design are listed in section 5 (changes).

9. Cross-group verification (Activity 3 Part D)

Project tested (group, repo)	Setup result	Roles exercised	Issues found	Feedback sent
Group __ - [TO COMPLETE]				

Feedback received on our project and fixes made: [TO COMPLETE]

10. Activity 4 - Database users, roles, privileges and access control

10.1 Projects set up

Group / project	Repository @ branch	Database	Objects (tables / views / sequences)	Setup result
Group __	[TO COMPLETE]			

10.2 Access-control plan

Reference plan, already implemented for our own project (`database/roles/`), which the group used as a model:

Login	Group role	Application role served	Responsibility	Privileges
rors_app	rors_runtime_grp	Application runtime	normal operations	S/I/U on customer, orders, payment; S/I/U/D on menu_item, restaurant_table, reservation, staff; S/I on order_item; sequences
rors_report	rors_readonly_grp	Viewer / auditor	reports	SELECT on non-credential tables and 4 views
rors_maint	rors_maint_grp	DBA - maintenance	VACUUM, ANALYZE, monitoring	pg_maintain, pg_monitor
rors_backup	rors_backup_grp	DBA - backup	pg_dump	pg_read_all_data
rors_dba	rors_admin_grp	DBA - administrator	administration	ALL on public tables/sequences, CREATE on schema

Per other project: one table like this in `db-access-control/group-XX/access-control-plan.md` [TO COMPLETE - copy the key rows here].

10.3 SQL scripts summary

`db-access-control/_template/00-05*.sql` (inventory, roles and grants, passwords by prompt, tuning, test matrix, privilege review) and `run-activity4.ps1`. Roles are prefixed per project (`g03_...`) because roles are cluster-wide. Folder link: [TO COMPLETE].

10.4 Verification results

Project	Account	Statement	Expected	Actual	Evidence
Group __	g__app	SELECT on authorised table	Allowed	[TO COMPLETE]	
Group __	g__report	UPDATE	Denied		
Group __	g__app	CREATE DATABASE	Denied		
Group __	g__app	connect to an unrelated database	Denied		

Reference result on our own database (`04_verify_privileges.sql`): [TO COMPLETE - paste PASS/FAIL summary].

10.5 Issues and fixes · 10.6 Feedback to owners

Project	Unexpected result	Cause	Fix / feedback sent (date)
Group __	[TO COMPLETE]		

11. Activity 5 - Database maintenance and monitoring

11.1 Projects monitored

Project	Repository	Database	Accounts and application roles used
Group __	[TO COMPLETE]		g__maint (DBA-maintenance), g__report (viewer)

11.2 Health and activity evidence · 11.3 Log findings

Per project: output of `01_health_checks.sql` with interpretation, and `log-findings.md` (evidence, explanation, impact, next step; or what was checked if nothing was found). [TO COMPLETE]

11.4 Maintenance performed

Project	Command	Why chosen	Account / date	Result
Group __	`ANALYZE;` and `VACUUM (VERBOSE, ANALYZE) public.<table>;`	refresh planner statistics; reclaim dead tuples; no VACUUM FULL (exclusive lock)	g__maint	[TO COMPLETE]

11.5 Query performance

Project	Query (application role)	Scan type	Est. vs actual rows / time	Recommendation
Group __	[TO COMPLETE]			

11.6 Recommendations, limitations, feedback

Project	Priority	Recommendation	Limitation (permissions/logging)	Feedback sent
Group __	Immediate / Follow-up / None	[TO COMPLETE]		

12. Activity 6 - Backup, restoration and recovery

12.1 Projects covered and backup plan

Project	Database	Account	Method	Storage	Schedule / retention	RPO / RTO
Group __	[TO COMPLETE]	g__backup	pg_dump -Fc	protected folder outside the repository	nightly / 7 daily, 4 weekly, 3 monthly	e.g. 24 h / 2 h

Roles after restore: `pg_dump` of one database does not contain cluster-wide roles or the database-level CONNECT grant, so the Activity 4 script is re-run for the restored database and one permitted and one denied operation are tested.

12.2 Backup creation and inspection

Project	File name	Size	SHA-256	Date/time	`pg_restore --list` evidence
Group __	[TO COMPLETE - from evidence/backup-log.csv]				

12.3 Restoration and verification · 12.4 Access control after restore

Project	Restore database	Objects	Row counts equal	Constraints present	Access control re-created and tested	Result
Group __	`<db>_restore_test`	[TO COMPLETE]				

12.5 Recovery simulation

Project	Incident (disposable row)	Timeline	Method	Recovered?
Group __	[TO COMPLETE]		fresh restore into `<db>_recovery`, copy the row back	

Loss after the newest backup would be unrecoverable from that file - this is what the RPO expresses.

12.6 Evaluation, limitations and errors, feedback

Is nightly `pg_dump` enough for a restaurant system? [TO COMPLETE - discuss RPO/RTO measured vs proposed; WAL archiving/PITR, off-site copies, restore drills]

13. AI usage declaration

AI tool	Tasks performed	How the group validated or corrected the output
Claude	Generated the application code, SQL scripts, role scripts, Activity 4-6 templates and the documentation skeleton from the approved Activity 2 design	[TO COMPLETE - what was run, what failed, what was changed]
ChatGPT / Gemini	[TO COMPLETE if used]	

Members must be able to explain the triggers, the exclusion constraint, the GRANT/REVOKE structure and the backup/restore commands. No real passwords, API keys or secrets appear in this document.